2nd Semester 2024-2025 TST

Paly Math Club

January 15, 2025

This test is the team-selection test for **Carnegie Mellon Informatics and Mathematics Competition** and **Stanford Math Tournament**.

CMIMC	SMT
Date: March 15, 2025	Date: April 12, 2024
Cost: $600 - $700	Cost: $0
Location: CMU, Pittsburgh, PA	Location: Stanford, Palo Alto, CA

Please circle the competitions you are planning to attend (This decision is **binding**):

1) CMIMC 2) SMT

Instructions: This is a **60** minute test with **15** questions. Each question is given a one point, and you will be ranked based on your total number of points. No point deduction for incorrect answers, no partial credit.

Name: _____

1. Michael has 16 white socks, 3 blue socks, and 6 red socks in a drawer. He randomly takes socks from the drawer (one at a time) until he has a pair of the same color. What is the largest number of socks he could possibly withdraw in this fashion?

2. The corner of a unit cube is chopped off such that the cut runs through the three vertices adjacent to the vertex of the chosen corner. What is the height of the cube when the freshly-cut face is placed on a table?

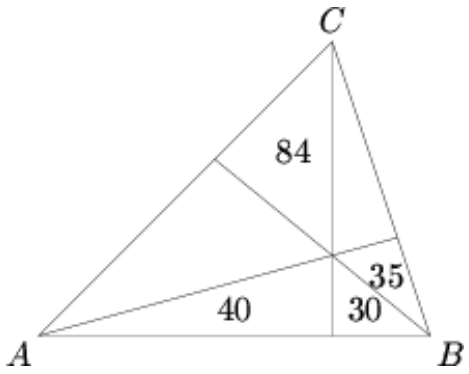
3. What is the sum of the solutions to the equation $\sqrt[4]{x} = \frac{12}{7-\sqrt[4]{x}}$?

4. What is the probability that a randomly chosen positive divisor of 10^{99} is an integer multiple of 10^{88}?

5. Brian has a 20-sided die with faces numbered from 1 to 20, and George has three 6-sided dice with faces numbered from 1 to 6. Brian and George simultaneously roll all their dice. What is the probability that the number on Brian's die is larger than the sum of the numbers on George's dice?

6. When a certain biased coin is flipped five times, the probability of getting heads exactly once is not equal to 0 and is the same as that of getting heads exactly twice. What is the probability that the coin comes up heads in exactly 3 out of 5 flips?

7. As shown in the figure, $\triangle ABC$ is divided into six smaller triangles by lines drawn from the vertices through a common interior point. The areas of four of these triangles are as indicated. Find the area of $\triangle ABC$.



8. A gardener plants three maple trees, four oaks, and five birch trees in a row. He plants them in random order, each arrangement being equally likely. What is the probability that no two birch trees are next to one another?

9. Let a, b, and c be the 3 roots of $x^3 - x + 1 = 0$. Find $\frac{1}{a+1} + \frac{1}{b+1} + \frac{1}{c+1}$.

10. Call a positive integer N a *7-10 double* if the digits of the base-7 representation of N form a base-10 number that is twice N. For example, 51 is a 7-10 double because its base-7 representation is 102. What is the largest 7-10 double?

11. The Fibonacci sequence is defined by the recurrence

$$F_n = \begin{cases} 1 & \text{if } n = 1 \text{ or } n = 2, \\ F_{n-1} + F_{n-2} & \text{otherwise.} \end{cases}$$

Find

$$\sum_{n=1}^{\infty} \frac{F_n}{4^{n+1}}$$

12. For each permutation $a_1, a_2, a_3, \cdots, a_{10}$ of the integers $1, 2, 3, \cdots, 10$, form the sum

$$|a_1 - a_2| + |a_3 - a_4| + |a_5 - a_6| + |a_7 - a_8| + |a_9 - a_{10}|.$$

The average value of all such sums can be written in the form $\frac{p}{q}$, where p and q are relatively prime positive integers. Find $p + q$.

13. Calvin has a bag containing 50 red balls, 50 blue balls, and 30 yellow balls. Given that after pulling out 65 balls at random (without replacement), he has pulled out 5 more red balls than blue balls, what is the probability that the next ball he pulls out is red?

14. The sides of rectangle $ABCD$ have lengths 10 and 11. An equilateral triangle is drawn so that no point of the triangle lies outside $ABCD$. The maximum possible area of such a triangle can be written in the form $p\sqrt{q} - r$, where p, q, and r are positive integers, and q is not divisible by the square of any prime number. Find $p + q + r$.

15. Let $f(x) = 2x^3 - 3x$. For what positive values of a do there exist distinct points b, c, d such that $(a, f(a)), (b, f(b)), (c, f(c)), (d, f(d))$ is a rectangle?